



Kapil Mulay
Bachelor of Technology
in Engineering Physics
Indian Institute Of Technology, Ropar

+91-9552730673
2024epb1266@iitrpr.ac.in
GitHub | Website
linkedin.com/in/kapil-mulay-b16335245

EDUCATION

Degree	Institute/Board	CGPA/Percentage	Year
Bachelor of Technology	Indian Institute of Technology, Ropar	9.16 (Till 4th Sem)	2024-2028
Senior Secondary	Central Board of Secondary Education	88.17%	2024
Secondary	Central Board of Secondary Education	96.2%	2022

EXPERIENCE

- **Dhananjay Lab for Education Design [Certificate]**

Junior Developer under Dr. Sudarshan Iyengar

– Implemented a research paper on background authenticity detection using Python, CNNs, and traditional feature extraction techniques.

– Trained a custom model for real vs. fake background classification; converted the model to TFLite for deployment.

– Built a frontend interface using React, Vite, and TypeScript to enable real-time inference in the browser.

– Integrated YOLO and MediaPipe for object detection and pose estimation as part of the vision pipeline.

– Packaged the full-stack application for distribution using Inno Setup.

November 2024 - January 2025

IIT Ropar
- PROJECTS
- **Maze Solver – Autonomous Maze-Navigating Micromouse**

Python, C++, ROS2, ESP32

– Implemented a Flood-Fill based navigation system in Python and C++ for autonomous maze solving with dynamic wall updates and real-time rerouting.

– Designed a multi-run traversal strategy (explore, return, optimize) with loop removal, diagonal path compression, and linear regression-based feasibility scoring to select the fastest path.

April 2025 - September 2025

GitHub | Simulation

• **NeuroGAN – ML Based Medical Image Synthesis Platform**

PyTorch, FastAPI, Next.js, Hugging Face Spaces, GitHub Actions

– Developed a DCGAN-based system trained on 10,000+ brain MRI images for synthetic tumor image generation.

– Implemented latent-space interpolation (LERP/SLERP) with an interactive interface for exploring GAN latent spaces.

– Built and deployed a full-stack application using FastAPI, Next.js, Hugging Face Spaces, and Vercel with automated CI/CD.

January 2026 - May 2026

GitHub | Demo

• **HydroSense – Cloud-Native IoT Water Quality Monitoring System**

TypeScript, Next.js, Node.js, MQTT, MongoDB Atlas, Prisma, HiveMQ Cloud

– Engineered an end-to-end IoT platform for water quality monitoring, ingesting real-time sensor data through MQTT.

– Built a cloud-native data pipeline with HiveMQ Cloud, MongoDB Atlas, and Prisma for persistent storage and analytics.

– Integrated ML-based water potability prediction and automated deployment on HF Spaces through GitHub Actions CI/CD.

January 2026 - May 2026

GitHub | Website

• **Grabbit – Full-Stack E-Commerce Platform**

Laravel, MySQL, Vite, PHP

– Implemented user authentication, inventory management, and order processing workflows.

– Designed a relational database schema in MySQL to manage users, products, orders, and transaction records.

– Integrated RazorPay gateway and automated emails to support secure checkout, order confirmations, and user notifications.

September 2022 - July 2023

GitHub | Website
- TECHNICAL SKILLS
- **Programming Languages:** C/C++, Python, JavaScript, TypeScript, PHP, Solidity

• **Tools & Frameworks:** Linux, Git, GitHub, Colab, FastAPI, PyTorch, NextJS, Laravel, Hardhat, ROS2
- KEY COURSES TAKEN
- **CSE & Maths:** Algorithm & Data Structures, Linear Algebra, Advanced Calculus, Probability and Stochastics

• **Electrical:** Signals and Systems, Digital Circuits, Control Theory
- POSITIONS OF RESPONSIBILITY
- **Assistant Internship Coordinator (SDE)**, Career Development and Placement Cell, IIT Ropar

May. 2026 - May. 2027

• **Deputy Contingent Lead**, Representing IIT Ropar at Inter IIT Tech Meet 15.0

May. 2026 - Feb. 2027

• **Coordinator, AI Club & Software Community**, Board of Science and Technology, IIT Ropar

May. 2025 - May. 2026

• **UG Mentor**, ISMP, IIT Ropar

May. 2025 - May. 2026
- MISCELLANEOUS
- **Top 7% Merit**, IIT Ropar

2025

Received the Top 7% Merit Prize from the institute in 2nd and 3rd semesters.

• **Inter IIT 14.0**, IIT Patna

2025

Secured 10th and 11th places in ISRO(AI) and Adobe problem statements. Represented IIT Ropar at the Engineer's Conclave.

• **Technoxian 2025**, Noida

2025

Represented IIT Ropar in the maze-solver challenge at Technoxian 2025 held in Noida Stadium.